Physics-Informed Neural Network for Heat Transfer in 2-D Homogeneous Material Systems and Complex Geometry

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***Abstract*—This paper presents a Physics-Informed Neural Network (PINN) based framework for modelling transient two-dimensional heat transfer in homogeneous material systems for two computational domains: a solid square and a square with a central void. The first domain is a solid square plate of side length 0.02 m, while the second is a square plate of side length 0.03 m containing a central void of side length 0.01 m. In both cases, the temperature field varies with spatial coordinates and time, which makes the problem a transient heat conduction problem. ANSYS-generated datasets containing node coordinates, time values, and temperature values in Kelvin are used for training and evaluation. The datasets are preprocessed through cleaning, normalization of spatial and temporal inputs, train-test partitioning, and masking of the void region for geometry-aware prediction analysis. A fully connected feedforward neural architecture is trained using a combined loss formulation that includes data fidelity and physics consistency. The model is implemented using TensorFlow, trained for 3000 epochs with the Adam optimizer, and assessed using mesh visualization, convergence curves, scatter plots, temperature distribution plots, and absolute error maps. The results show that the proposed PINN framework is able to learn the overall transient thermal behaviour in both a regular conductive domain and a more complex domain with an internal geometric exclusion. The study highlights the usefulness of PINNs as a mesh-free, scalable, and physically guided approach for thermal modelling.**

***Index Terms*—Physics-Informed Neural Networks, Heat Transfer, Transient Conduction, ANSYS, Deep Learning, Complex Geometry, Thermal Modelling**

I. Introduction

Heat transfer plays a central role in engineering because temperature distribution directly affects efficiency, safety, reliability, and system performance. Accurate modelling of transient thermal behaviour is necessary in electronics cooling, material processing, energy systems, thermal protection structures, and industrial heat-management applications. Since temperature depends on both space and time, many practical engineering systems require transient rather than steady-state modelling.

Traditional methods such as Finite Element Analysis (FEA), Finite Difference Methods (FDM), and Finite Volume Methods (FVM) are widely used for solving thermal problems governed by partial differential equations. These methods are reliable and accurate, but they require discretization, mesh generation, repeated numerical solves, and careful treatment of boundary conditions and geometry. The difficulty increases further when the domain contains internal cutouts, voids, or other geometric irregularities.

Physics-Informed Neural Networks (PINNs) offer a different way to solve such problems. Instead of learning only from labelled data, a PINN also learns from the governing physical equation. This gives the neural network both data guidance and physics guidance. Since PINNs use coordinate inputs directly, they provide a mesh-free approximation framework and can act as surrogate models after training.

In this work, a PINN-based framework is applied to transient two-dimensional heat conduction in two domains:

1) a solid square plate of side length 0.02 m,
2) a square plate of side length 0.03 m with a central square void of side length 0.01 m.

The temperature data for both domains are generated in ANSYS and exported into Excel files. The model uses spatial coordinates and time as input and predicts temperature as output. The main objective is to examine whether a physics-guided neural network can accurately learn the temperature evolution for both a regular conductive domain and a geometrically interrupted conductive domain.

The major contributions of this work are:

- development of a PINN framework for transient 2-D heat transfer,
- study of both a simple solid square and a square-with-void geometry,
- use of ANSYS-generated datasets for supervised and physics-guided training,
- detailed evaluation through convergence curves, scatter plots, temperature distributions, absolute error maps, and cross-domain comparison.

II. Literature Review

A. Overview of PINNs

Physics-Informed Neural Networks have emerged as an important approach in scientific machine learning because they integrate governing physical equations directly into the training process. In a purely data-driven model, the objective is

only to fit observed outputs. In contrast, a PINN also penalizes violations of the differential equation that governs the system. This allows the learned model to be not only numerically accurate but also physically meaningful.

PINNs are useful when the system is described by a known equation and the solution field varies continuously over space and time. Because the network directly maps coordinates to field values, it behaves as a continuous function approximator. This property makes PINNs attractive in fluid mechanics, structural mechanics, heat transfer, and inverse modelling.

B. PINNs in Heat Transfer

Heat transfer is a strong candidate for PINN-based modelling because thermal systems are governed by diffusion equations. In transient heat conduction, temperature fields evolve smoothly through time, and this makes the problem suitable for learning through a combination of data and physics. PINNs can therefore be applied to forward conduction problems, inverse estimation of thermal parameters, and surrogate modelling for repeated evaluations.

C. Role of Geometry

Complex geometry remains one of the major challenges in computational heat transfer. Internal holes, excluded regions, or sharp changes in domain shape complicate conventional numerical workflows because they require careful meshing. The square-with-void domain used in this study is a useful example of this challenge. Although PINNs avoid mesh dependence during prediction, geometry still needs to be handled carefully in preprocessing and interpretation. In this work, masking is used so that the central void is excluded from the final thermal visualization.

III. Geometry and Physical Problem Formulation

Two two-dimensional domains are considered in this work. The first is a solid square plate of side length 0.02 m, with:

$$x, y \in [-0.01, 0.01] \text{ m}. \tag{1}$$

The second is a square plate of side length 0.03 m containing a central square void of side length 0.01 m. The outer domain is:

$$x, y \in [-0.015, 0.015] \text{ m}, \tag{2}$$

and the void region is:

$$x, y \in [-0.005, 0.005] \text{ m}. \tag{3}$$

The thermal behaviour is governed by the transient two-dimensional heat equation:

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right), \tag{4}$$

where T is the temperature field and α is the thermal diffusivity. In this work:

$$\alpha = 9.7 \times 10^{-5} \text{ m}^2/\text{s}. \tag{5}$$

The thermal setup applies a temperature difference across the plate, with the west side maintained at 500 K and the east side at 0 K, while the north and south boundaries are treated as adiabatic. These conditions generate a left-to-right thermal gradient and a transient conduction process.

IV. ANSYS Dataset Description and Analysis

A. Dataset Generation

The datasets were generated using ANSYS simulations for the two domains. The ANSYS solution provides node-level temperatures at multiple time instants, which are exported to spreadsheet format for learning and evaluation. The files used in the project are:

- `square.xlsx`,
- `void.xlsx`.

Each row in the dataset represents one spatial point at one time instant and contains node number, spatial coordinates, time, and temperature.

B. Dataset Size and Structure

The solid square dataset contains 7695 points, while the square-with-void dataset contains 3600 points. The time values span from 1 s to 15 s. Therefore, the learning task can be represented as:

$$(x, y, t) \to T, \tag{6}$$

where the neural network takes normalized coordinates and time as input and predicts temperature in Kelvin as output.

The solid square dataset is larger because the entire region is conductive and sampled continuously. The void dataset is smaller because the excluded central region reduces the number of valid conductive points. This difference in size also affects the density of information available during training and slightly changes how each model learns the thermal field.

C. Physical Meaning of the Data

The solid square dataset represents continuous heat diffusion across a homogeneous conductive plate. The square-with-void dataset captures heat transfer in a domain where the central region is removed from conduction. This leads to a different thermal pattern because heat must redistribute around the void rather than passing through the center.

This difference between the datasets is important because it affects both learning difficulty and interpretation of results. The solid square is expected to show smoother and more uniform gradients, while the void case is expected to show geometry-induced redistribution of temperature. Thus, the two datasets do not merely represent different numbers of points; they represent different physical pathways of heat flow.

D. Preprocessing of the Dataset

The preprocessing pipeline includes:

1) loading the Excel datasets,
2) selecting spatial coordinates, time, and temperature,
3) removing inconsistent or missing entries,
4) normalizing the spatial coordinates and time,
5) splitting the data into training and testing subsets,

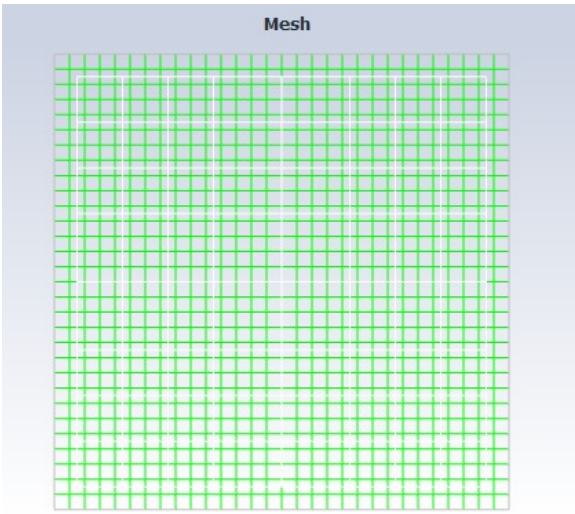


Fig. 1: Mesh for the solid square domain.

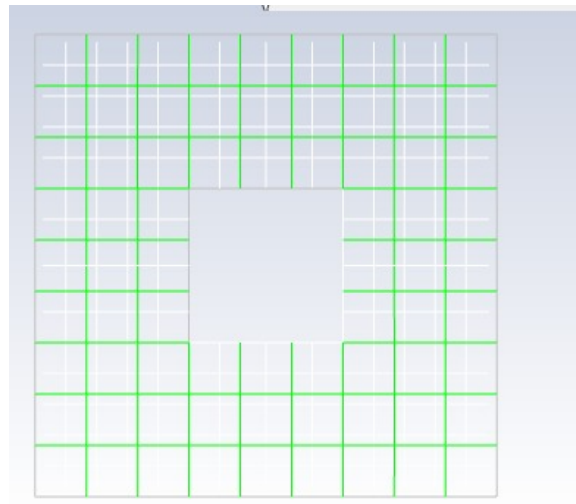


Fig. 2: Mesh for the square-with-void domain.

6) applying a mask for the void geometry during visualization.

The normalization is performed using Min-Max scaling:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \tag{7}$$

$$y' = \frac{y - y_{\min}}{y_{\max} - y_{\min}}, \tag{8}$$

$$t' = \frac{t - t_{\min}}{t_{\max} - t_{\min}}. \tag{9}$$

Temperature is retained in Kelvin so that the output remains physically interpretable. A 70:30 train-test split is used. For the void case, masking ensures that the central empty region is not displayed as if it contains valid thermal values. These preprocessing steps are essential because they bring the data into a stable numerical range, maintain physical meaning in the output, and preserve the geometric characteristics of the problem.

V. Methodology

A. Neural Network Architecture

The predictive model is a fully connected feedforward neural network. Its input is:

$$(x, y, t), \tag{10}$$

and its output is:

$$\hat{T}(x, y, t). \tag{11}$$

The network architecture used in this work consists of:

- 64 neurons,
- 32 neurons,
- 16 neurons,
- 8 neurons,
- 1 output neuron.

ReLU activation is used in the hidden layers. The architecture is chosen to balance expressive power and computational efficiency.

This layered design allows the network to progressively transform the input coordinates into a meaningful temperature prediction. The larger early layers help the model capture broad nonlinear spatial-temporal relationships, while the later narrower layers compress the learned features into a refined scalar temperature output. In other words, the architecture acts as a hierarchy of feature extraction and feature refinement. The first hidden layer helps the model learn coarse thermal tendencies, the middle layers refine the relationship between location, time, and temperature, and the final small hidden layer helps map those learned features into a physically meaningful temperature estimate.

The choice of a fully connected architecture is also suitable for this project because the problem is coordinate-based. Each point in the domain is represented only by x, y, and t, and the objective is to learn a continuous mapping from these variables to temperature. Since the output field is smooth and governed by diffusion, the feedforward architecture provides a reasonable and computationally manageable approximation framework.

B. Loss Formulation

The total loss combines data consistency and physics consistency. The data loss is:

$$\mathcal{L}_{data} = \frac{1}{N} \sum_{i=1}^{N} (\hat{T}_i - T_i)^2. \tag{12}$$

The residual of the governing heat equation is:

$$r(x, y, t) = \frac{\partial \hat{T}}{\partial t} - \alpha \left(\frac{\partial^2 \hat{T}}{\partial x^2} + \frac{\partial^2 \hat{T}}{\partial y^2} \right), \tag{13}$$

and the physics loss is:

$$\mathcal{L}_{physics} = \frac{1}{N} \sum_{i=1}^{N} r_i^2. \tag{14}$$

The total loss is then:

$$\mathcal{L}_{total} = \mathcal{L}_{data} + \mathcal{L}_{physics}. \tag{15}$$

The purpose of this combined loss is to ensure that the network does not merely memorize thermal samples. Instead, it learns a solution that remains close to both the dataset and the heat-transfer physics. The data term anchors the model to the ANSYS reference solution, whereas the physics term discourages predictions that violate the transient diffusion equation. This dual guidance is one of the most important aspects of the PINN framework.

C. Training Strategy

Both models are trained for 3000 epochs using the Adam optimizer with learning rate 0.001. During training:

1) the normalized inputs are passed through the network,
2) the predicted temperatures are obtained,
3) automatic differentiation computes the required derivatives,
4) the total loss is evaluated,
5) gradients are backpropagated,
6) model parameters are updated.

The solid square case trains over a continuous conductive region. The square-with-void case uses the same learning principle but requires geometry-aware masking in the final analysis. Even though the same training pipeline is used for both domains, the void case is structurally more challenging because the thermal field must reorganize around the excluded region. Therefore, this shared training strategy becomes useful for comparing how the same model framework behaves under different geometric conditions.

VI. Experimental Setup

The implementation uses Python and TensorFlow, with NumPy for numerical operations, pandas for spreadsheet handling, and Matplotlib for visualization. The main configuration settings are:

- architecture: $64 \to 32 \to 16 \to 8 \to 1$,
- activation: ReLU,
- optimizer: Adam,
- learning rate: 0.001,
- epochs: 3000,
- train-test split: 70:30.

Evaluation is carried out through:

- convergence graphs,
- predicted-versus-actual scatter plots,
- time-based temperature distributions,
- absolute error maps,
- cross-domain comparative interpretation.

The temperature distribution and error maps are examined at 1 s, 5 s, 10 s, and 15 s. These time instants are sufficient to reveal both early-stage and later-stage transient behaviour. By selecting multiple points across the conduction timeline, the analysis can explain how the model behaves during sharp early gradients and during smoother later diffusion.

VII. Results

A. Training Convergence

The convergence curves show how the total loss changes across epochs. In both domains, the loss falls sharply in the early stage of training and then gradually stabilizes. This indicates that the network quickly learns the dominant thermal pattern and then uses later training epochs for refinement.

The solid square convergence plot suggests rapid learning of the regular thermal gradient. The void case also converges strongly, which indicates that the network can handle additional geometric complexity without unstable optimization. The similar overall behaviour of the two convergence curves shows that the training process is numerically stable in both domains, even though the thermal pathways are different.

B. Scatter Plot Analysis

The scatter plots compare predicted temperatures with ANSYS reference values on the test set. A good model should produce points clustered closely around the diagonal line $y = x$, which indicates agreement between prediction and target.

The scatter plots indicate strong agreement in both domains. The reported correlation is approximately 94% for the solid square and 95% for the square-with-void case. This suggests that the trained PINN is able to reproduce the thermal field accurately across unseen data points.

These scatter plots are important because they complement the convergence curves. The convergence plots show that the loss is decreasing, but the scatter plots show what that decrease means in practical predictive terms. The strong alignment around the diagonal confirms that the reduction in loss translates into meaningful agreement with the ANSYS temperature field, not just numerical optimization progress.

C. Temperature Distribution and Error Analysis for the Solid Square

The solid square results show the predicted temperature field and the corresponding absolute error map at four time instants.

At 1 s, the gradient is sharper because the thermal field has not yet fully redistributed. At later times, the field becomes smoother, which is consistent with transient diffusion in a homogeneous plate. The error maps show that deviations remain bounded and structured, with slightly stronger values near regions of higher gradient.

Another important observation is that the thermal field remains globally smooth across the domain at all four time instants. This shows that the network is not producing unstable or noisy thermal patterns. Instead, it maintains the expected physical continuity of a homogeneous conductive plate. This is a useful sign that the learned model is respecting the dominant structure of the diffusion process.

D. Temperature Distribution and Error Analysis for the Square with Void

The square-with-void results reveal how the thermal field develops around the excluded central region.

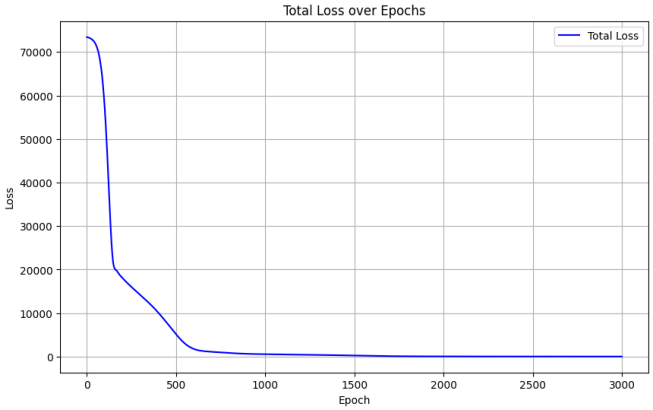


Fig. 3: Total loss over epochs for the solid square model.

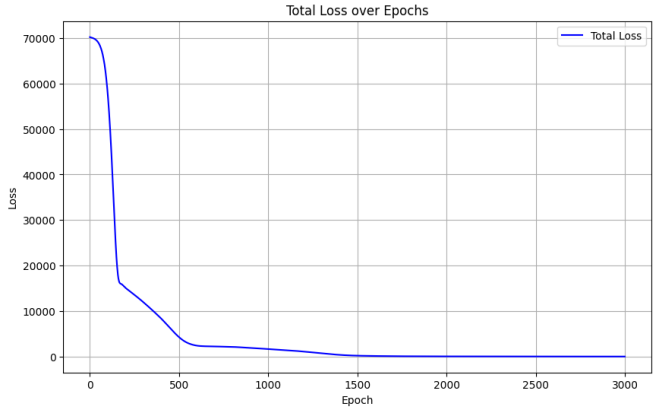


Fig. 4: Total loss over epochs for the square-with-void model.

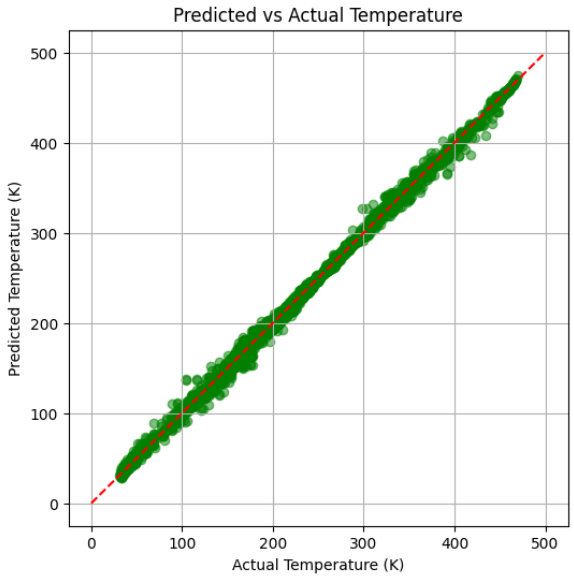


Fig. 5: Predicted versus actual temperature for the solid square domain.

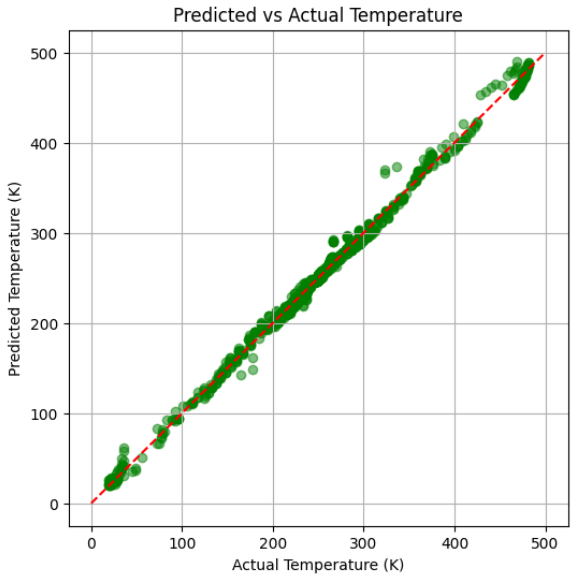


Fig. 6: Predicted versus actual temperature for the square-with-void domain.

The temperature field bends around the void instead of passing through it, which is physically correct. The masked central region remains blank, confirming that the geometry-aware post-processing is handled properly. The error maps remain low over most of the conducting region, although sharper boundary effects and geometry-induced changes make some local regions harder to predict.

These results are especially important because they show that the model is learning more than a simple left-to-right thermal gradient. It is also learning the geometric redistribution effect produced by the void. In other words, the network is not only fitting temperature values but also reproducing the way the domain shape alters the spatial pattern of heat transfer.

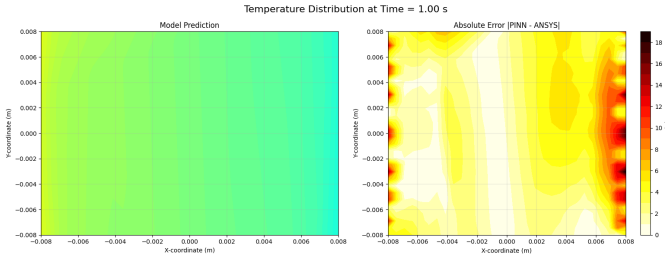
E. Comparative Analysis of the Two Domains

A direct comparison between the solid square and square-with-void cases provides important insight into the behaviour of the proposed PINN framework.

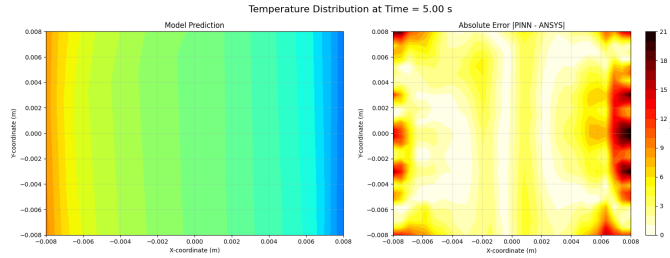
First, the solid square is geometrically simpler and provides a continuous conductive path across the full domain. Because of this, the temperature gradient is smoother and more uniform. The network therefore learns a more regular mapping from (x, y, t) to temperature, which leads to stable convergence and low, spatially smooth error patterns.

Second, the square-with-void case is geometrically more complex because the internal void interrupts the conduction path. This changes the heat-flow pathway and forces the thermal field to redistribute around the excluded region. As a result, the learning problem becomes more challenging in a spatial sense, even though the global temperature pattern remains structured. The model still performs strongly, which indicates that the PINN framework adapts well to geometry changes when masking is used correctly.

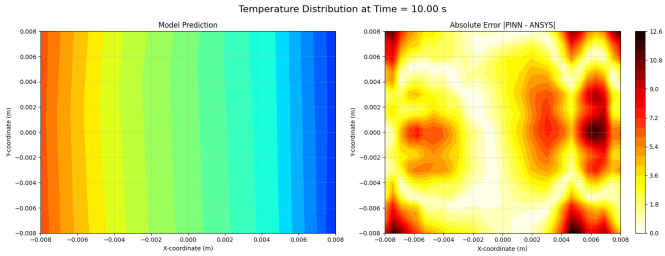
Third, the convergence behaviour in both domains shows that the network learns efficiently. The square-with-void case does not show unstable behaviour despite the added complexity. This suggests that the chosen architecture and loss formu-



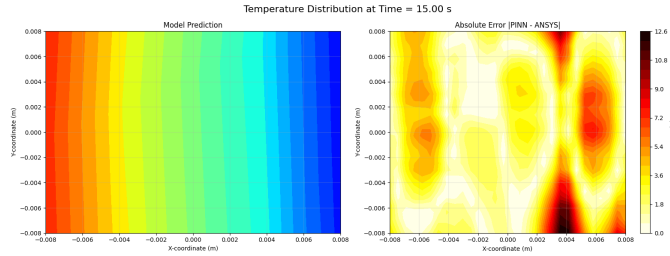
(a) 1 s



(b) 5 s

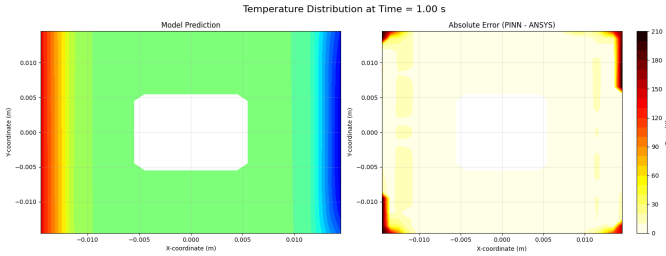


(c) 10 s

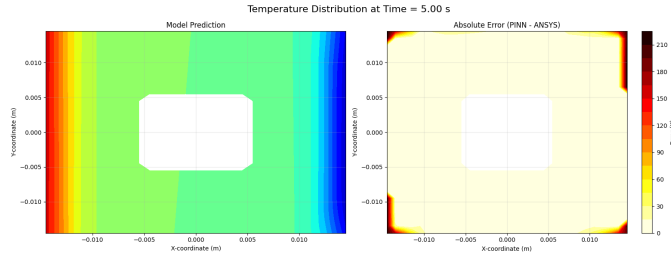


(d) 15 s

Fig. 7: Predicted temperature field and absolute error maps for the solid square domain at four time instants.



(a) 1 s



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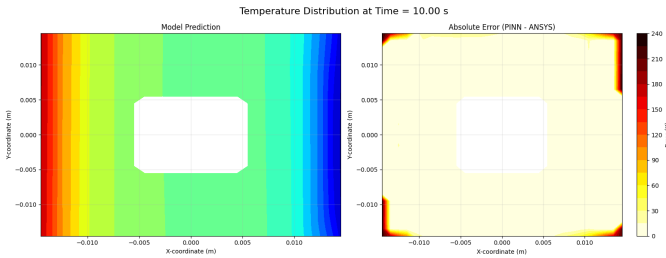

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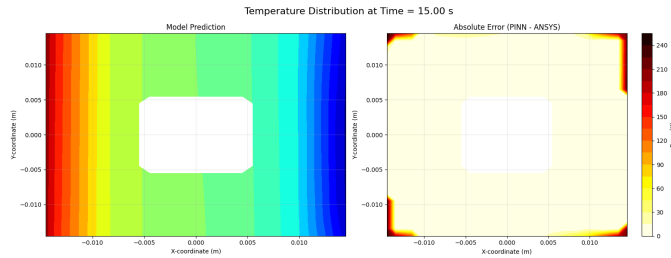

(d) 15 s

Fig. 8: Predicted temperature field and absolute error maps for the square-with-void domain at four time instants.

lation are sufficiently robust for both regular and interrupted conductive domains.

Fourth, the scatter plots indicate strong agreement in both cases, with the square-with-void domain showing slightly higher reported correlation. This may be due to the constrained nature of the conductive region, which reduces some forms of internal variation while preserving a strong boundary-driven thermal structure.

Finally, the temperature distributions and error maps show the real difference between the two domains. In the solid square, the model mainly learns a smooth left-to-right transient gradient. In the square-with-void case, the model must learn both the global thermal gradient and the local redistribution effect caused by the void. That the network succeeds in both cases is one of the main findings of this work.

F. Overall Interpretation

Taken together, the convergence curves, scatter plots, distribution plots, error maps, and direct comparison show that the PINN learns both global and local thermal behaviour successfully. The solid square demonstrates performance in a regular conductive domain, while the square-with-void case demonstrates that the same framework can remain effective even when the geometry becomes more complex.

This overall interpretation also helps explain why the framework is useful as a surrogate modelling approach. It learns the broad thermal trend, respects the geometric characteristics of the domain, and produces interpretable outputs across multiple

evaluation views. This combination of qualities is important when the goal is not just numerical prediction, but meaningful physical modelling.

VIII. Conclusion and Future Work

A. Summary of the Work

This study presented a Physics-Informed Neural Network framework for transient two-dimensional heat transfer in homogeneous material systems. Two domains were examined: a solid square and a square with a central void. ANSYS-generated temperature data were used to train neural models that take spatial coordinates and time as inputs and predict temperature as output.

The work included dataset preparation, normalization, geometry-aware handling, PINN-based modelling, and detailed visual evaluation. The framework combines supervised learning with physical consistency, which makes it more meaningful than a purely data-driven regression model.

B. Key Findings

The results show that the proposed framework can learn the overall thermal behaviour in both regular and geometrically interrupted domains. The solid square case demonstrates strong performance in a continuous conductive region. The square-with-void case shows that the same framework can also handle a more complex geometry when the void is treated properly in visualization and interpretation.

The convergence plots indicate stable training, the scatter plots show strong agreement with the ANSYS data, the time-based thermal distributions are physically interpretable, and the error maps show that the remaining deviations are localized and meaningful.

C. Limitations

The present work is limited to two-dimensional homogeneous materials and simulation-generated data. The final performance therefore depends strongly on the quality and coverage of the ANSYS datasets. The study also emphasizes visual and correlation-based interpretation more than a broad set of numerical error metrics.

D. Future Work

Several directions can improve and extend the current work:

- stronger and more explicit handling of boundary conditions,
- refinement of the physics-loss formulation,
- extension to heterogeneous materials,
- extension to three-dimensional heat transfer,
- comparison with conventional numerical methods in terms of accuracy and runtime,
- development of faster surrogate models for practical engineering use.

Future work can also explore more advanced neural architectures and broader datasets so that the framework becomes more robust for practical heat-transfer prediction tasks.